CBAM and the CAP

Towards a coherent climate framework for agriculture

Bipin Joshi and group members **Albrecht Daniel Thaer-Institute · 22. July 2026**

Table of Content

1. Why did the EU create ETS and CBAM

2. Impacts on International Markets :

- Farners and Markets
- EU and Agricultural sector

3. Sustainability Integration in Agriculture



1.Why did the EU create the ETS and CBAM ?

How does the EU actually put a price on carbon?

Global context : climate goals

- Climate neutrality by 2050 : net-zero greenhouse gas emissions
- -55% Gas House Emissions by 2030 (vs. 1990 levels)
- **ETS + CBAM** as the EU’s core carbon-pricing tools

54%

- Of free ETS allowances (2021-25) go to sectors now covered by CBAM

The leakage problem

- EU producers pay for carbon, foreign competitors often don't
- Risk: production **relocates** outside the EU
- Result: emissions are moved abroad, not reduced globally

1. How does the ETS tool works ?

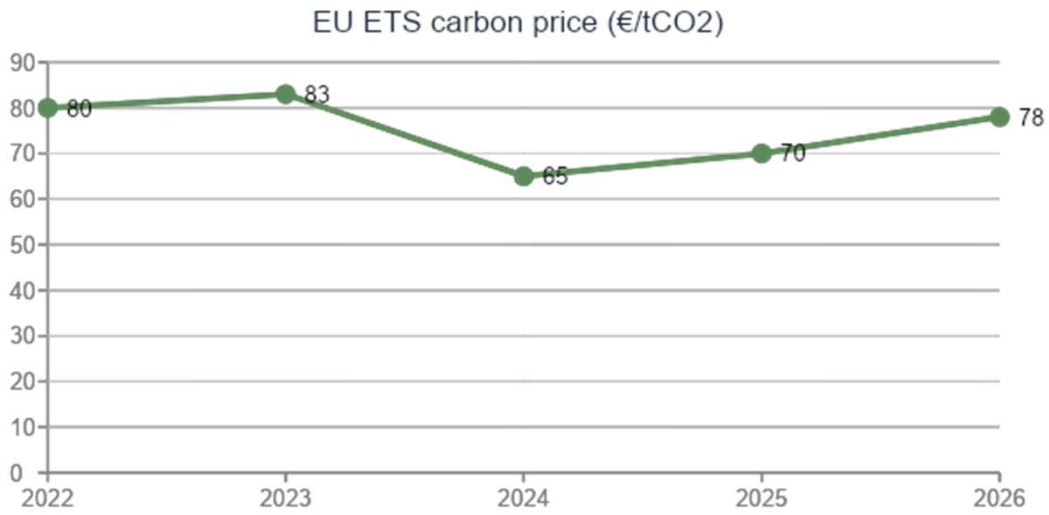
A market where the shrinking emissions cap does the heavy lifting, not the price tag itself.

Cap-and-trade mechanics

- EU sets a total emissions cap, in place since 2005
- Companies receive or buy allowances (EUAs)
- Surplus allowances can be resold on the market

-4.3%/year

Reduction rate (Linear Reduction Factor) of the EU-wide emissions cap



The ETS turns a legal, shrinking cap into a real market price for carbon
So the stronger the price, the stronger the pressure to cut emissions.

European Environment Agency. (2025). *Use of auctioning revenues generated under the EU Emissions Trading System*. https://www.eea.europa.eu/en/analysis/indicators/use-of-auctioning-revenues-generated

European Securities and Markets Authority. (2024, October 7). *ESMA market report: EU carbon markets 2024*(ESMA50-43599798-10379). https://www.esma.europa.eu/sites/default/files/2024-10/ESMA50-43599798-10379_Carbon_markets_report_2024.pdf

1. From Free Allowances to Border Pricing

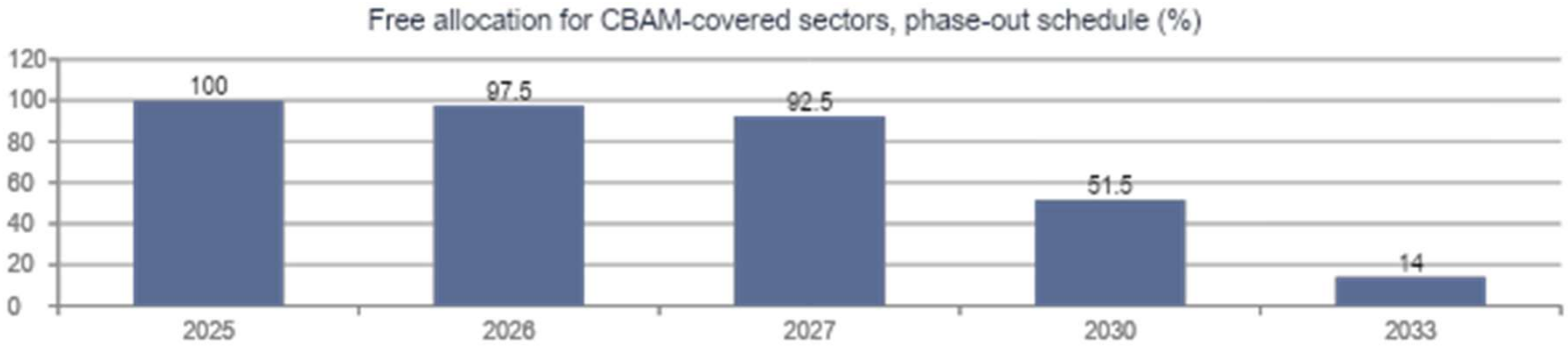
Same goal, two tools : internal protection gradually giving way to external pricing.

Free allowances

- Historically shielded exposed sectors: steel, cement, fertilisers
- Reduced their incentive to decarbonise
- Being phased down: -2.5% (2026), -5% (2027)

CBAM : rising protection

- Charges importers the same carbon cost as EU producers
- Phased in exactly as free allocation phases out
- Definitive regime in force since January 2026



As free allowances fade out, CBAM becomes the EU's new shield against carbon leakage.

European Parliament, Legislative Train Schedule. (n.d.). *Carbon border adjustment mechanism as part of the European Green Deal*, from https://www.europarl.europa.eu/legislative-train/theme-a-european-green-deal/file-carbon-border-adjustment-mechanism

Regulation (EU) 2023/956 of the European Parliament and of the Council of 10 May 2023 establishing a carbon border adjustment mechanism. Official Journal of the European Union

1. CBAM : Carbon Border Adjustment Mechanism

Sectors covered by CBAM

- Cement, steel, aluminium
- Hydrogen, electricity
- No general coverage of agriculture : only Fertilizers

January 2026

start of the definitive CBAM regime, meaning real certificate purchases, priced around €75/tCO2

Focus : Fertilizers

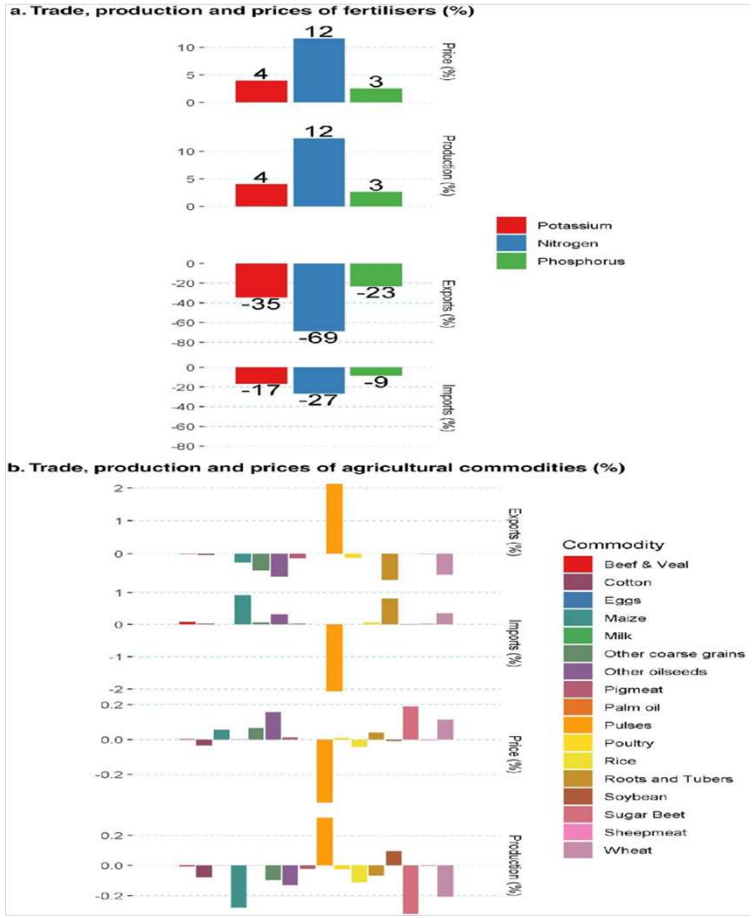
- The one CBAM sector that is a direct farm input
- Higher import costs for fertilisers = higher costs for farmers
- Ripple effects: farm costs, food prices, trade flows



→ ***Impacts on Agriculture: Farmers, Food Market, Trade***

Impact on Farmers and Markets (I)

How will CBAM affect the fertilizer industry, the agricultural sector and consumer prices?



Fertilizer and Chemical industry – A level playingfield

- Clear guidelines necessary for longterm investments (§27a poses an uncertanty)
- Innovation boost towards greener fertilizer production and systems Lead markets

Are farmers squeezed in the middle?

- No margin(s) for error: Already tight businesses are hit hardest by geopolitical developments
- Call for suspension & direct support
- Reluctant willingness to adapt

Sources: Pieralli et al. (WILEY-BLACKWELL 2025); EURACTIV Panel ‚The EU Fertilizer Action Plan…' 9.06.2026; Copa- Cogeca: Ref. COMM(26)00296[1]

Impact on Farmers and Markets (II)

What policy measures has the Commission envisioned to stabilize the market short-term ?
How do the industry and farmers adapt and innovate medium-term?

Fertilizer Action Plan	**CAP Adjustment** **Financial relief** **Value Chain Partnership**	• Liquidity scheme under rural development • 540 €million + national top ups • Open dialogue and problem solving
‚Greener' Production	**GMO's/Legouminous plants** **CO2 free production e.g. with bacteria** **Circular Production**	• Technological Innovation & natural fertilizing • Biotech-Startups • Sludge Reuse and Biomethane

Sources: EURACTIV Panel ‚The EU Fertilizer Action Plan…' 9.06.2026; Baudry, Ravelonjanahary (Chaire Économie du Climat, Working paper 11/2025); EU Comission Press Release, ‚Comission proposes 540 million in stress relief`` 12.06.2026; ‚Dünger der Zukunft: Die Stickstoff Revolution' (Arte 2026); ‚How Farmers can cut emissions' (Financial Times 2024)

Impact on International Markets

Incentivizing green transition beyond EU market

Emission Accounting (Li, 2026)

- Default emission valued higher than actual emissions
- Market incentivization for verifiable emission accounting
- Building block for national carbon markets

National Markets (Li, 2026)

- CBAM actualizes lost potential carbon market revenue
- Incentivization of national carbon markets

Unequal Burden (Lin, 2024)

- Economic and technological differences could lead to economic exclusion
- Unequal economic impact on Non-OECD countries
- No allocated funds to assist developing countries through CBAM transition

Geopolitical Tension

- Retaliatory measures- within unpredictable markets and current political landscape

Trade Distortion (GATT 1947)

WTO GATT Article XX Exceptions

(*b*) necessary to protect human, animal or plant life or health

(*g*) relating to the conservation of exhaustible natural resources if such measures are made effective in conjunction with restrictions on domestic production or consumption;

CBAM incentivizes Non-EU nations towards implementing carbon markets

EU and International Agricultural Sector

Agricultural exceptionalism at within the EU and beyond

EU Competition (DY, 2025)

- Environmental policy can be critiqued as negatively impacting the EU competition
- Preserving competition by internalizing carbon cost at an international level

Agricultural Exceptionalism

- Agricultural exceptionalism
- Agricultural sector emissions are not explicitly under CBAM or ETS
- Environmental sustainability within the agricultural market focuses on incentives ‘carrots’ over penalties ’sticks’

Agriculture Environmental Impact (Stepanyan, 2026)

- Agriculture sector is large contributor to global ghg emissions
- Fertilizer is taxed under CBAM
- CBAM showcases ‘sticks’ method to decrease emissions within the agricultural sector

CBAM preserves EU competition while addressing environmental sustainability through penalties

CBAM Climate Impact

Can CBAM deliver on GHG emission reduction goals?

EU Emissions (OECD, 2025)

ETS functioning with CBAM

Agricultural emission reduction

Global Emissions (OECD, 2025)

Global ghg emissions are expected to fall overall and within the agriculture sector

CBAM and Global Decarbonization

- Market push towards less carbon intensive technologies
- Market push towards national carbon markets
- Potential CBAM revenue

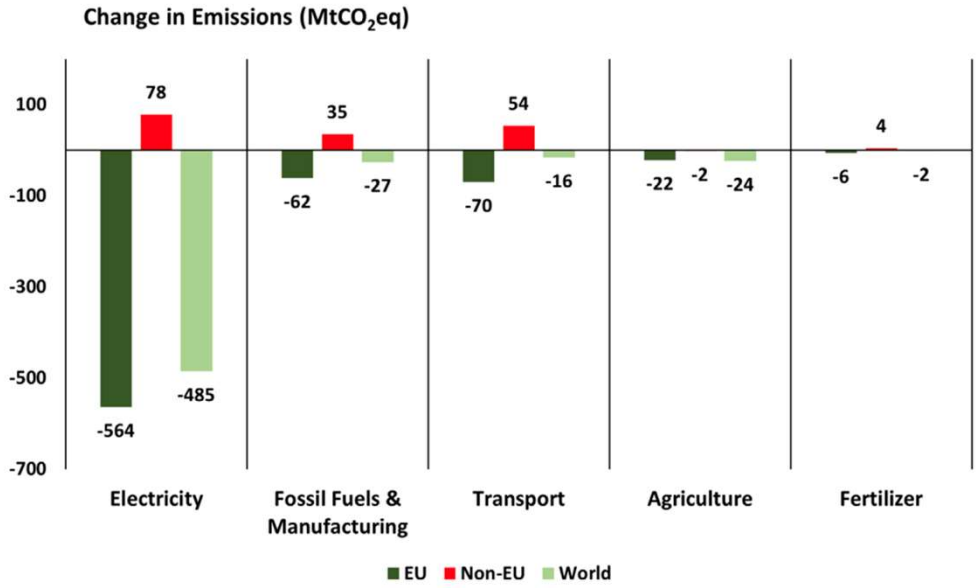


Figure : Absolute change in emissions in 2030 for economic sectors compared to the reference scenario **(Stepanyan, 2026)**

CBAM is predicted to prevent carbon leakage and lower global CO2 emissions

Current Developments / Overall Timeline

EU Emissions Trading System (EU ETS)
2005

European Green Deal
2019

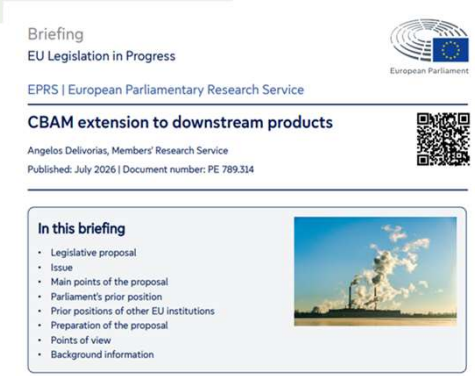
CBAM proposal
July 2021

CBAM Regulation
May 2023

Transitional phase
2023 – 2025

Beginn of CBAM
January 2026

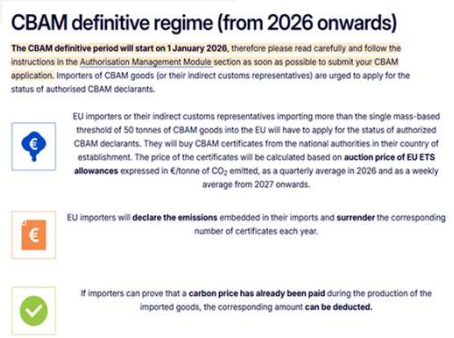
Cbam Extensions
December 2025


Briefing
EU Legislation in Progress
EPRS | European Parliamentary Research Service
CBAM extension to downstream products
Angelos Delivorias, Members' Research Service
Published: July 2026 | Document number: PE 789.314
In this briefing
- Legislative proposal
- Issue
- Main points of the proposal
- Parliament's prior position
- Prior positions of other EU institutions
- Preparation of the proposal
- Points of view
- Background information

https://www.europarl.europa.eu/RegData/etudes/BRIE/2026/789314/EPRS_BRI(2026)789314_EN.pdf

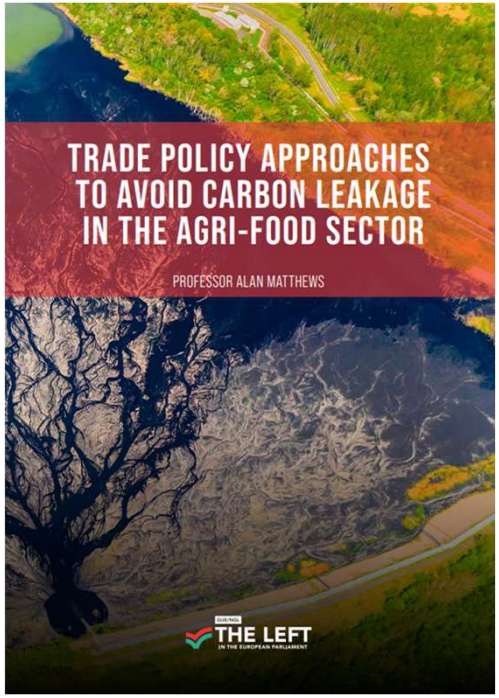
CBAM-Omnibus
September 2025

CBAM definitive regime


CBAM definitive regime (from 2026 onwards)
The CBAM definitive period will start on 1 January 2026, therefore please read carefully and follow the instructions in the Authorisation Management Module section as soon as possible to submit your CBAM application. Importers of CBAM goods (or their indirect customs representatives) are urged to apply for the status of authorised CBAM declarants.
EU importers or their indirect customs representatives importing more than the single mass-based threshold of 50 tonnes of CBAM goods into the EU will have to apply for the status of authorized CBAM declarants. They will buy CBAM certificates from the national authorities in their country of establishment. The price of the certificates will be calculated based on **auction price of EU ETS allowances** expressed in €/tonne of CO_2 emitted, as a quarterly average in 2026 and as a weekly average from 2027 onwards.
EU importers will **declare the emissions** embedded in their imports and **surrender** the corresponding number of certificates each year.
If importers can prove that a **carbon price has already been paid** during the production of the imported goods, the corresponding amount **can be deducted.**

https://ec.europa.eu/commission/presscorner/detail/en/ip_25_3088

CBAM and the Agricultural Sector (I)



Trade Policy Approaches to Avoid Carbon Leakage in the Agri-food Sector (2022)

The European Union's Carbon Border Adjustment Mechanism: Challenges and Prospects for the Agricultural Sector (2025)



Pascal Xicluna/agriculture.gouv.fr

19 mai 2025 Info +

The European Union's Carbon Border Adjustment Mechanism: Challenges and Prospects for the Agricultural Sector – Analysis n°207

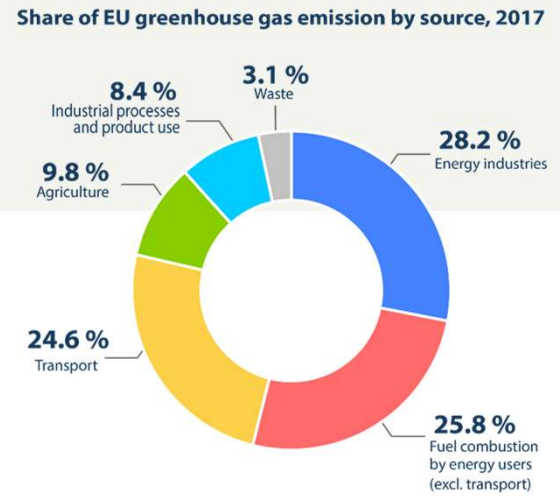
CEP english contents changement climatique décarbonation engrais compétitivité commerce international prospective

https://agriculture.gouv.fr/european-unions-carbon-border-adjustment-mechanism-challenges-and-prospects-agricultural-sector#section-5;
https://left.eu/app/uploads/2023/02/GUE-Study-TRADE-Carbon-leakage.pdf

CBAM and the Agricultural Sector (II)

The central challenge is not extending CBAM, but aligning climate ambition with fair competition and sustainable trade

- Agriculture contributes around 10% of the EU's greenhouse gas emissions
- Imported products such as beef or dairy can have high carbon footprints
- Equal carbon pricing could create a more level playing field



- Agricultural emissions (especially methane and nitrous oxide) are much harder to measure than industrial CO_2 emissions
- Production systems vary widely across countries
- It could increase food prices
- It may disproportionately affect developing-country exporters
- WTO and food security concerns remain

https://ec.europa.eu/eurostat/cache/digpub/energy/2019/bloc-4a.html

Sustainability Integration in Agriculture (I)

While there's no CBAM for agriculture, sustainability is already being built into farming through the CAP.

Is the EU building one coherent sustainability system for agriculture — inside the farm gate and at its borders — or several disconnected instruments?

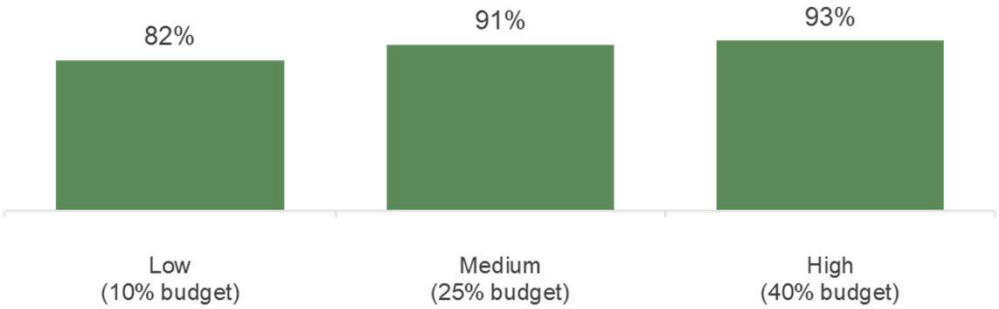
Enhanced Conditionality — Mandatory baseline

- Set entry conditions for direct payments — replaced the old "greening" rules.
- Farm-model evidence (Petsakos et al., 2023) : conditionality— drives most of the environmental gain, at a modest income cost (≈ 2–3.5%).

Eco-schemes (Pillar I) — Voluntary layer

- Voluntary for farmers (for going beyond minimum rules) — MS must ring-fence at least 25% of Pillar 1 direct payments. Highly heterogeneous: 3–21 measures per country.
- Large farms and horticulture farms adopt least, since foregone income can exceed the payment.

Evidence: eco-scheme uptake responds to how much budget is at stake



Eco-scheme adoption rises with budget ambition (high-conditionality scenario, EU-wide farm model)

Farm to Fork targets supported by the CAP

25%	−50%	−20%	10%
farmland organic	pesticide risk	fertiliser use	biodiversity features on Farm land

Gap: CAP Biodiversity requirements remain below the 10% biodiversity target, and climate payments are not consistently linked to measurable emission reductions.

CAP already promotes sustainable agriculture through mandatory rules and voluntary incentives, but there are still gaps before the EU has a fully coherent agri-climate system. And also the CAP mostly looks at what happens inside the farm.

Sources: Cuadros-Casanova et al. (2023, Conservation Biology); Runge et al. (2022, EuroChoices); Petsakos et al. (2023, Applied Economic Perspectives and Policy)

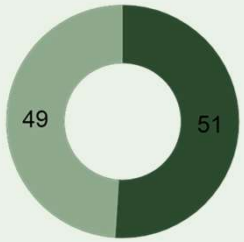
Sustainability Integration in Agriculture (II)

Sustainability requirements now extend beyond farm payments to trade and carbon markets.

EUDR — Deforestation-free supply chains

- **Covers:** cattle, cocoa, coffee, palm oil, rubber, soy, wood — goods must be deforestation-free (post-2020), legal, and due-diligence checked.
- **Timeline:** large/medium operators by Dec 2026, small operators by June 2027 (postponed twice; compliance costs cut ~75%)
- **Critique:** a unilateral, demand-side tool — risks just shifting deforestation elsewhere without matching action in producer countries

Why EUDR matters



Just 3 countries —
Brazil, Colombia & Indonesia — accounted for 51% of global tropical primary forest loss in 2022.

EUDR applies uniformly EU-wide, even though deforestation risk is geographically concentrated.

Carbon farming & the CRCF framework

- **The EU's first certification framework** for voluntary carbon farming — in force since December 2024
- **Rewards farmers** for building up soil carbon, and links to CAP support — rolling out through 2026.
- **Caution**: 'sequestration' often just means slower losses, not real carbon removal — measuring long-term storage is scientifically hard.

CRCF "QU.A.L.ITY" quality criteria – represent real climate benefits

Quantification	Additionality
Long-term storage	Sustainability

EU sustainability policies now extend beyond domestic farming to global supply chains and carbon markets. However, differences between instruments like EUDR and CRCF show that these policies are still developing separately rather than forming a fully integrated framework.

Sources: Muradian et al. (2025, Ecological Economics); Paul et al. (2023, Journal of Environmental Management); European Commission / Council of the EU (Regulation (EU) 2024/3012; Regulation (EU) 2025/2650)

Connections Between CBAM & CAP

Bringing CBAM back in how the border mechanism and the CAP fit together.

CAP

Internal transition

Pays EU farmers to adopt lower-emission practices; operates through subsidies, not a carbon price — agricultural emissions stay outside the EU ETS.

EU Green Deal / Farm to Fork

CBAM

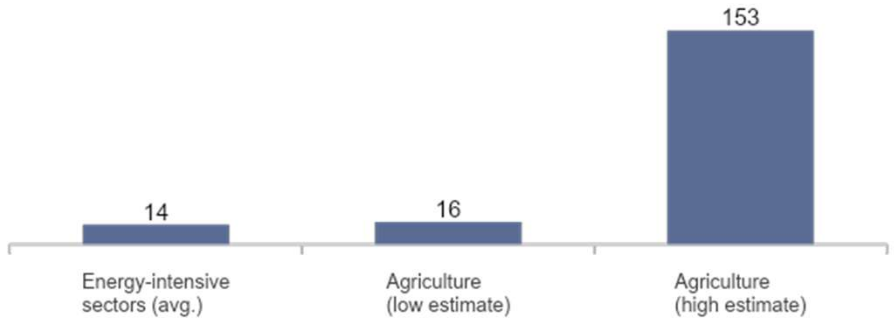
External protection

CBAM places a carbon cost on imports to prevent carbon leakage and ensure EU producers facing carbon prices are not disadvantaged. Currently covers cement, steel, fertilizers, hydrogen, and electricity — not agriculture.

The missing link: why there is no Ag-CBAM (yet)

- CBAM requires a domestic carbon price to adjust import costs and prevent carbon leakage.
- Agriculture is regulated through CAP payments and conditionality, not carbon pricing, so there is currently no carbon cost to border-adjust.
- Future carbon pricing and CRCF developments could make an Ag-CBAM possible to protect EU farmers' competitiveness

Why agriculture may need this protection most



A rate above 100% means emissions rise abroad by more than they fall in the EU.

Source: Fournier Gabela et al. (2024, Climate Policy)

Sources: Fournier Gabela et al. (2024, Climate Policy); Cuadros-Casanova et al. (2023, Conservation Biology)

Toward a Coherent EU Agri-Climate Framework?

Answering the guiding question: how far do CAP, EUDR, CRCF and CBAM add up to one system?

Instrument	Where it acts	How it works	Status (mid-2026)
CAP (conditionality + eco-schemes)	Inside the farm gate	Payments tied to environmental standards	In force 2023–2027; adoption uneven
EUDR	Import supply chains	Due-diligence, deforestation-free proof	Applies Dec 2026 (large/medium)
CRCF / carbon farming	Voluntary carbon market	Certified SOC & removal credits	In force; methodologies rolling out 2026
CBAM	EU border, priced sectors	Levy on embedded carbon of imports	Live for industry; no ag-CBAM proposed

Conclusion

Overall, the EU is gradually building a more integrated agri-climate framework. While the CAP supports the transition within agriculture, CBAM protects climate ambition at the border. Agriculture is not yet fully connected to this system, but current policy developments — CRCF, EUDR, rising CAP conditionality — suggest the for-and-against ag-CBAM debate may become harder to postpone.

Discussion: as agricultural sustainability requirements continue to increase, should border protection evolve alongside them?

Sources: Cuadros-Casanova et al. (2023, Conservation Biology); Runge et al. (2022, EuroChoices); Petsakos et al. (2023, Applied Economic Perspectives and Policy); Muradian et al. (2025, Ecological Economics); Paul et al. (2023, Journal of Environmental Management); Fournier Gabela et al. (2024, Climate Policy)

References (1/2)

- **Arte.** (2026). *Dünger der Zukunft: Die Stickstoff-Revolution*
- **Baudry, & Ravelonjanahary.** (2025). *Chaire Économie du Climat, Working Paper No. 11/2025.*
- **Copa-Cogeca.** (2026). *Reference COMM(26)00296.*
- **Cuadros-Casanova et al.** (2023). *Conservation Biology.*
- **European Commission.** (2025). *Commission proposes €540 million in stress relief for farmers.* https://ec.europa.eu/commission/presscorner/detail/en/ip_25_3088
- **European Environment Agency.** (2025). *Use of auctioning revenues generated under the EU Emissions Trading System.* https://www.eea.europa.eu/en/analysis/indicators/use-of-auctioning-revenues-generated
- **European Parliament.** (2026). *Carbon Border Adjustment Mechanism: Implementation and latest developments* (EPRS Briefing No. 789314). https://www.europarl.europa.eu/RegData/etudes/BRIE/2026/789314/EPRS_BRI(2026)789314_EN.pdf
- **European Parliament.** (n.d.). *Carbon border adjustment mechanism as part of the European Green Deal.* Legislative Train Schedule. Retrieved July 21, 2026, from https://www.europarl.europa.eu/legislative-train/theme-a-european-green-deal/file-carbon-border-adjustment-mechanism
- **European Securities and Markets Authority.** (2024). *EU carbon markets 2024* (ESMA50-43599798-10379).
- **Eurostat.** (2019). *Energy statistics – Greenhouse gas emissions.* https://ec.europa.eu/eurostat/cache/digpub/energy/2019/bloc-4a.html
- **EURACTIV.** (2026). *The EU Fertilizer Action Plan* [Panel discussion, June 9, 2026].
- **Financial Times.** (2024). *How farmers can cut emissions.*
- **Fournier Gabela et al.** (2024). *Climate Policy.*
- **French Ministry of Agriculture and Food Sovereignty.** (n.d.). *The European Union's Carbon Border Adjustment Mechanism: Challenges and prospects for the agricultural sector.* https://agriculture.gouv.fr/european-unions-carbon-border-adjustment-mechanism-challenges-and-prospects-agricultural-sector
- **Muradian et al.** (2025). *Ecological Economics.*
- **Paul et al.** (2023). *Journal of Environmental Management.*
- **Petsakos et al.** (2023). *Applied Economic Perspectives and Policy.*
- **Pieralli et al.** (2025). *Wiley-Blackwell.*
- **Regulation (EU) 2023/956 of the European Parliament and of the Council of 10 May 2023 establishing a Carbon Border Adjustment Mechanism.** (2023). *Official Journal of the European Union.*
- **Regulation (EU) 2024/3012 of the European Parliament and of the Council.** (2024). *Official Journal of the European Union.*
- **Regulation (EU) 2025/2650 of the European Parliament and of the Council.** (2025). *Official Journal of the European Union.*
- **Runge et al.** (2022). *EuroChoices.*
- **The Left in the European Parliament.** (2023). *Trade policy approaches to avoid carbon leakage in the agri-food sector.* https://left.eu/app/uploads/2023/02/GUE-Study-TRADE-Carbon-leakage.pdf

References (2/2)

- Dy, K. B., & Yang, H. (2025). Global emissions and international trade: Quantifying the impact of carbon border adjustments. Energy Economics, 145, 108373. https://doi.org/10.1016/j.eneco.2025.108373
- GATT 1947, World Trade Organization, https://www.wto.org/english/docs_e/legal_e/gatt47_e.htm
- Li, M., Sun, Y., Xia, Y., Yang, Z., Chen, C., Yuan, Y., & Wang, P. (2026). "The impacts of Carbon Border Adjustment Mechanism (CBAM) on international trade and policy responses: from an economic and environmental equity perspective." *Energy Policy*, 210(C), Elsevier. https://www.sciencedirect.com/science/article/pii/S030142152500521X
- Lin, B., & Zhao, H. (2024). Threatening the Poor? The economic impacts of carbon border adjustment mechanism on developing countries. Structural Change and Economic Dynamics, 71, 582–593. https://doi.org/10.1016/j.strueco.2024.09.005
- OECD (2025). "EU Carbon Border Adjustment Mechanism: What is it, how does it work and what are the effects?" https://www.oecd.org/en/blogs/2025/03/eu-carbon-border-adjustment-mechanism-what-is-it-how-does-it-work-and-what-are-the-effects.html
- Pieralli, S., Pérez-Domínguez, I., Elleby, C., & Cluff, M. (2025). Implications of the EU's carbon border adjustment mechanism for fertilizer and food markets. EuroChoices, 24, 23–30. https://doi.org/10.1111/1746-692X.12469
- Stepanyan, D., Zamani, O., Heidecke, C., Pelikan, J., Rieger, J., Osterburg, B., & Gocht, A. (2026). "Fertilising climate policy: the dual impact of CBAM on EU agricultural emissions and regional disparities." Journal of Agricultural Economics, May 2026. https://literatur.thuenen.de/digbib_extern/dn071146.pdf

Thank You!